

Evidence-bounded encounter forecasting: an honest-model architecture for effort-biased wildlife observation data

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Abstract

Wildlife encounter forecasts are typically presented as smooth probability maps that carry no record of the evidence that produced them. Observers, stewards, and field researchers cannot determine whether a displayed hotspot reflects animal behavior, observer effort, acoustic detector false positives, or model overfitting. orcast is a pilot platform for evidence-bounded encounter forecasting on southern resident killer whales in the Salish Sea that enforces a different contract: the forecast is always shown, but its displayed confidence is bounded by automated statistical gates, explicit integrity conditions, per-cell provenance traces, quarantined citizen-science ingest, and a human promotion step, all persisted in DynamoDB with an auditable chain from map cell to source record. A grounding benchmark comparing structured artifact-bound citation (prepare-then-narrate with step-log annotation) against Google Maps place-grounded generation on the same marine science query finds that place grounding leaves 85% of scientific evidence claims uncited while orcast annotations bind every claim to a producing skill, artifact identifier, and data reference. The system is deployed at <https://orcast-h0.vercel.app> and the grounding benchmark is reproducible via the provided measurement tool.

1 The four gaps in encounter forecasting

Wildlife encounter forecasting produces maps of predicted animal presence or detection probability. In practice these products present four compounding failures.

The **epistemic gap** is that a map shows an intensity or probability without exposing what evidence earned it. Users of AI-assisted spatial forecasts systematically overtrust displayed confidence when the basis of that confidence is not made visible alongside the prediction [126]. Survey practitioners have documented that uncertainty is routinely suppressed in data-driven maps even when authors know it exists [120]. The problem is not that maps are untrustworthy; it is that the contract between evidence quality and displayed certainty is not explicit.

The **effort gap** is that sightings and acoustic detections conflate animal presence with observer presence. Movement ecology models show that the classical mass-action approximation fails when observer search effort is spatially structured [32]. In cetacean monitoring, shore observers concentrate near access points, ferries, and interpretable coastlines, so naive hotspot aggregation learns where people watch, not where whales travel. The machine learning wildlife conservation literature acknowledges this as a central unresolved problem in field data pipelines [33].

The **provenance gap** is that no existing operational system exposes a per-prediction drilldown from a displayed probability to the kernels, detections, and gate verdicts that produced it. A researcher cannot audit whether a displayed confidence reflects a fitted diel kernel that passed a phase-shuffled null test and a held-out deviance skill check, or a prior-dominated estimate with no model fit.

The **governance gap** is that citizen-science shore reports and automated model promotions enter operational forecasting narratives without explicit quarantine or signed human authority. The information quality literature on citizen science establishes that unmoderated crowdsourced data produces systematic quality failures when used directly in probabilistic models [17]. The Gravity Spy program at LIGO provides a concrete operational precedent: citizen classifier outputs are quarantined and validated before influencing detector characterization [207].

Existing tools address pieces of this. No operational system combines live forecast, gate battery, per-cell provenance, quarantined community ingest, and immutable promotion audit in one loop backed by a primary transactional database.

2 A gate-bounded honesty model

2.1 Distributional family

Encounter counts are sparse, overdispersed, and zero-inflated. The negative binomial distribution arises as a gamma-Poisson marginalization and provides two free parameters to capture overdispersion and random effects [272]. Hurdle and NB models are required for species abundance estimation when Gaussian assumptions fail on ecological count data [13]. Species distribution modeling frameworks identify structured noise and under-counting as reasons that count-specific distributional families are

necessary [75].

The kernel model uses the log-linear intensity, decomposing the log encounter rate into a baseline plus additive cyclic kernel contributions from diel, lunar, and seasonal covariates:

$$\log \lambda(x, t) = b_0 + \sum_k \beta_k \phi_k(t) + \log E(x, t) \quad (1)$$

See also the methodology of record.

2.2 Null gating

Each fitted cyclic kernel must beat a phase-shuffled null before contributing to displayed confidence. The phase-shuffled null generates a reference distribution by permuting the assignment of detection times to stimulus phases. Point process models for spike trains [59] and the time-elapsed neural model [58] establish the mathematical foundation.

$$p_{\text{null}} = P(T(\text{shuffled}) \geq T(\text{observed})) \quad (2)$$

The time-rescaling goodness-of-fit test uses the compensator transformation to produce residuals that should be exponentially distributed; a KS test on these residuals is the second gate component [84]:

$$\tau_i = \Lambda(t_i) - \Lambda(t_{i-1}), \quad \tau_i \stackrel{H_0}{\sim} \text{Exp}(1) \quad (3)$$

An open measurement gap (C-2): this methodology is documented in neuroscience and statistical literature but lacks an ecology-domain citation in the arXiv corpus. The whitepaper cites methodological precedents.

2.3 CV skill and calibration

Gates L2 and L3 require positive held-out deviance skill

$$S_D = 1 - \frac{D_{\text{model}}}{D_{\text{null}}} \quad (4)$$

and PIT calibration

$$u_i = F_{\text{model}}(y_i), \quad u_i \stackrel{H_0}{\sim} \text{Uniform}(0, 1) \quad (5)$$

Negative skill means the model is less predictive than the null and displayed confidence must be withheld. The epidemiological calibration literature establishes that PIT-based recalibration reliably improves log score [102]. The abstention loss framework [119] formalizes the principle that a predictive system should withhold confidence when skill is low. Spatial and temporal CV must account for autocorrelation [107, 105].

2.4 Effective confidence gate

Displayed effective confidence is bounded by the product of gate outcomes:

$$c_{\text{eff}} = c_{\text{raw}} \cdot \prod_k g_k, \quad g_k \in \{0, 1\} \quad (6)$$

Failed gates reduce confidence and expose the failing integrity condition alongside the metric. The gate is not a visibility switch; the forecast is always shown.

3 Acoustic grounding: PSTH and Level 0 QC

Passive acoustic monitoring is the primary temporal instrument in orcast because acoustic detectors run continuously and their effort does not track human activity cycles.

3.1 PSTH kernel estimation

The PSTH method aligns detections to a repeating stimulus onset (diel sunrise, flood-current onset, lunar new moon) and bins by phase. The binned counts, divided by station effort and smoothed with a periodic spline, produce a first kernel estimate. Point-process models for spike trains [59] and the time-elapsed neural model [58] establish the Poisson-process machinery; the hydrophone is a detector unit, the detection times are the spike train, and the environmental cycles are the stimulus.

3.2 False-positive rates and quarantine

Deep learning bioacoustic classifiers deployed in field conditions exhibit non-trivial false-positive rates. The computational bioacoustics review [21] documents this as a known, widespread problem: performance degrades in field conditions relative to held-out test sets. The North Atlantic right whale upcall detector [47] demonstrates this for a marine mammal application: a ResNet classifier trained on thousands of examples requires careful validation under variable acoustic conditions. Robust sound event detection in bioacoustic networks [56] characterizes ambient noise variability as the primary driver of detection unreliability. For orcast, unreviewed OrcaHello candidates cannot enter a model fit directly. The Level 0 QC gate quarantines them in a separate data stream, and the integrity condition exposed alongside every confidence-bearing surface reports explicitly that the acoustic data consists of unreviewed candidates with known false-positive rates. The effective confidence gate (Section 2) prevents confidence from rising above a threshold while this condition holds.

The integrity condition is part of the forecast, not a reason to hide it.

4 Governance: quarantine, authority, and audit

Shore-based sighting submissions enter orcast through a moderation queue that enforces quarantine before any submission can influence model outputs. The governance architecture has three components: quarantine by default, signed human authority, and an immutable decision record.

4.1 Why quarantine is necessary

The information quality literature frames citizen science as a user-generated content problem: quality management pipelines are a necessary condition for scientific usability [17]. The LIGO Gravity Spy program provides an operational precedent: citizen classifier votes are aggregated and validated against expert labels before influencing detector characterization [207]. The Milky Way Project demonstrates that unmoderated citizen labels alone produce systematic errors without ensemble aggregation and expert calibration [210]. Galaxy morphology classification confirms that hybrid human-ML pipelines outperform either component alone only when citizen data is filtered for quality [208].

4.2 Signed human authority

orcast separates automated gate eligibility from human promotion authority. A model fit that passes L1 through L3 fitness gates is eligible for promotion, but promotion requires a signed decision by a registered reviewer recorded in the decision-records DynamoDB table. The record includes reviewer identity, verdict, cited gates, and a rationale field. The record is immutable and auditable.

4.3 Moderation queue mechanics

Community shore submissions are stored in a pending state. Approval stamps attribution and a low reliability weight before the submission can influence the model. Private field journal entries remain private unless the author publishes them to the community queue, at which point they enter the same moderation pipeline.

Every public confidence surface reflects only validated and promoted data; the provenance chain is auditable to the specific reviewer decisions that authorized each data point.

5 Provenance grounding as a system

Scientific grounding requires that every displayed claim can be traced to its producing method, the experiment that validated that method, the data the experiment ran on, and the research that motivated the experimental design.

5.1 Prepare-then-narrate

The concierge dispatches only allow-listed skills from a versioned manifest. Each skill invocation is recorded as a step. After all skills complete, the LLM narrates the collected JSON context. RAG reduces factual hallucination in conversational systems [142]. CRAG [143] demonstrates that corrective retrieval reduces hallucination when retrieval fails. AGREE [149] shows that adapting LLMs to self-ground claims and generate citations reduces unsupported-claim rates. The LLM attribution survey [151] confirms that generative systems without attribution produce high rates of unauditable unsupported claims.

5.2 Step-log provenance

Every interaction persists an ordered `steps` array in `exploration_turns` as JSONB: a `resolve_agent` step recording the cast role id, version, and spec hash; `skill_invocation` steps with grounding references; and a `model_output` step with annotations. CamFlow whole-system provenance [186] establishes the requirements. The provenance and data differencing paper [188] demonstrates step-log sufficiency for reproducibility. Scientific ML workflow provenance in geoscience [189] directly addresses this class of pipeline.

5.3 Claim/method/experiment graph

The ProvenanceGraph component renders the step log as a two-tier visual graph:

$$G = (V, E), \quad V = \{\text{metric}, C, M, X, D, R\} \quad (7)$$

Claim nodes (C) are constructed from model output annotations; each C node names the annotation type, bound artifact or href, and a no-signal badge if unbound. Method nodes (M) are constructed from skill invocation steps; each M node names the skill, status, and lists `grounding_refs` (artifact or dataset references populated during skill execution). The C and M tiers are rendered in the live UI. The full C/M/X/D/R schema described in the contract (X = experiment/artifact, D = provenance citation, R = research citation) is stored in the step-log JSONB and is the intended rendering target; X, D, and R nodes currently appear as inline references within M rows rather than as separate visual graph nodes. SciERC [249] establishes the joint entity-relation extraction foundation. SciClaim [250] introduces the fine-grained C/M/X annotation schema with typed relational edges. MindSearch [255] demonstrates graph-of-queries decomposition at scale. MindMap [248] demonstrates that KG-grounded reasoning reduces LLM hallucination.

Claim nodes without an artifact or href binding render a no-signal badge. This is the honesty primitive the geospatial grounding baseline (Section 7) lacks.

6 Orchestrated managed agents with verified tools

6.1 Plan-then-execute and tier verification

The surface planner emits a `ui_intent` specifying panels and a `skill_plan`. The concierge dispatches exactly those skills. Skills not in the cast role's allowed list return a 400 `tier_blocked` before invocation. The model narrates only after all skills complete.

ReAct [177] establishes the canonical plan-act-verify pattern: constraining the action space to a defined tool set makes the verify step tractable and reduces hallucination. Internal representations predict tool selection hallucination before execution [30]. HaluAgent [165] demonstrates that a structured typed toolbox enables reliable operation from smaller models. APIGen [25] establishes that verifiable function calling is an automatable engineering property.

6.2 Skill catalog tiers

T0 skills are public; T1 skills require a geographic viewport; T2/T3 are keyed and blocked on the public route. Cast roles specify an explicit skill subset, model, and policy. Unknown skill IDs return 400. Four cast roles are deployed: `explore-guide-v1`, `surface-planner-v1`, `dossier-explainer-v1`, and `promotion-clerk-v1`.

6.3 Step logs as training substrate

The persisted step-log corpus is the data foundation for embedding-indexed retrieval over prior fits and provenance chains (W-RAG, chartered separately). The step log is not only an audit record; it is the substrate for the system to improve its own grounding quality over time. TRACE [29] demonstrates that knowledge-grounded reasoning chains constructed from structured context eliminate irrelevant retrieval noise in multi-hop queries; the orca step-log is this structure applied to skill-dispatch interactions.

7 Geospatial grounding limits for scientific evidence

7.1 Benchmark methodology

Gemini 3.5 Flash (Api-Revision 2026-05-20) with the `google_maps` tool at San Juan Islands coordinates (48.5465, -123.03) was measured across three query types: total citation count, citation type split (place vs. scientific/dataset), grounding density per thousand words, and the unsupported scientific claim rate:

$$R_{\text{uncited}} = \frac{|\{s \in S_{\text{sci}} : \nexists \text{ cited_name} \in s\}|}{|S_{\text{sci}}|} \quad (8)$$

The orca comparison used the same orca evidence query submitted to `/api/interactions/plan`, counting annotations with a bound artifact or href. The measurement tool is available at `tools/testing/maps_grounding_probe.py`.

7.2 Results

Across three Maps grounding queries, 0 of 25 citations resolved to scientific or dataset sources; all 25 resolved to Google Maps place pins. The evidence query produced 9 scientific sentences (NOAA fisheries data, Center for Whale Research census, Chinook salmon corridors, J/K/L pod hydrophone data); 8 of 9 (89%) were uncited. Averaged across evidence-bearing queries: $R_{\text{uncited}} = 85\%$. The orca result: 6 of 6 annotations were bound; $R_{\text{uncited}} = 0\%$.

7.3 Interpretation

This does not reflect a deficiency in the Gemini or Maps system; it reflects the structural scope boundary of geospatial grounding. Maps grounding is designed for place information and excels at that: the POI query achieved 8 citations in 229 words with 0 unsupported sentences. Place grounding cannot resolve scientific evidence claims to their origin data, method, or experiment. The LLM geospatial knowledge study [229] provides peer-reviewed evidence that LLM geospatial competence degrades substantially for knowledge not encoded as named places. A domain-specific provenance system is required for the evidence layer.

7.4 Reproducibility

Benchmark run: 2026-06-24. Model: `gemini-3.5-flash`. API revision: 2026-05-20. Coordinates: 48.5465, -123.03. Fully reproducible via `tools/testing/maps_grounding_probe.py` with a Gemini API key.

7.5 Extended benchmark: 8-scenario parallel RAG comparison

A follow-on 8-scenario parallel benchmark (`tools/testing/grounding_parallel_rag.py`, 2026-06-24) compared Maps-only grounding against five RAG-augmented variants

using live orca skill outputs as context. The hierarchy of R_{uncited} across conditions reveals a diagnostic structure:

Scenario	Context injected	R_{uncited}
S4 (surface planner step-log)	Orchestrated reasoning trace	0%
S8 (hotspot data)	Structured probability data	75%
S1 (Maps-only orca evidence)	None	60–91%
S2 (gate context)	Narrow gate data	100%

The surface planner step-log (Scenario 4) eliminates R_{uncited} entirely by transforming the query from an open-domain science query into a closed artifact-reference query: the model answers from the step-log rather than from world knowledge. This is not improved citation generation; it is query transformation. Gate context (Scenario 2) does not reduce R_{uncited} because the injected gate data does not cover the scientific claims Gemini independently generates. The hierarchy functions as a diagnostic for grounding architecture quality: context that binds the query to a complete reasoning trace eliminates uncited claims; narrowly structured context does not. This finding motivates a formal evaluation framework for AI reasoning systems (see companion paper, *Grounding quality measurement for orchestrated AI reasoning chains*, in preparation) [149, 143].

8 Limits and falsification

8.1 Current scope limits

The pilot covers one hydrophone station (Haro Strait, OrcaHello) over approximately 7.5 months. Diel and lunar kernels are fitted; tide, season, and salmon index kernels are in the leveled program but not yet fitted at L1. The spatial kernel is prior-dominated. Effective confidence is 0% promoted pending a reviewer promotion decision.

Two search families returned Partial verdicts. The effort bias family lacks a cetacean-specific effort-correction measurement in the arXiv corpus; the analogs [31, 33] support the methodological claim but not a cetacean-domain number. The null gating family lacks ecology-journal documentation of phase-shuffled null tests for diel/lunar covariates; the neuroscience precedents [59, 58] establish the statistical machinery.

8.2 Falsification table

The following conditions would weaken or falsify hypothesis H1 in this deployment:

Observation	Interpretation
Displayed confidence rises without a promotion decision	Gate policy failed; H1 weakened
Provenance modal omits a failed gate	Honesty contract violated
Citizen submission influences model without moderation approval	Corollary 3 falsified
Sighting check returns confident yes/no without uncertainty disclosure	prepare-then-narrate violated
New covariate passes gates but CV deviance skill is negative	Correct outcome; gate correctly withholds confidence

8.3 Extension path

Adding stations, reviewed OrcaHello labels, effort covariates, and spatial kernels re-runs the full gate battery. Confidence moves only when gates and human authority agree. The W-Eval study (chartered separately) tests H1 directly through a structured field observer panel comparing steward trust in the gate-bounded forecast against a confidence-smoothed baseline.

8.4 Connection to world model evaluation

The provenance and grounding infrastructure described here has a direct connection to a broader evaluation gap in AI systems that reason about

the physical world. LeCun’s autonomous machine intelligence architecture [15] specifies that world model intermediate representations must be grounded in their generating sensor observations, and that planning outputs must be evidence-bound. V-JEPA [3] and Image World Models [27] are trained and evaluated on downstream accuracy benchmarks; no claim-level evidence-binding metric is proposed. Physical-world AI benchmarks (PHYRE [2]; STAR [28]; ThreeDWorld [23]; TravelPlanner [269]) all measure task completion or planning success without auditing whether intermediate claims are bound to sensor observations. The Elements of World Knowledge framework [26] tests world modeling via plausibility matching, not evidence binding. The orcast grounding benchmark (R_{uncited} , Section 8) provides the evaluation primitive this literature lacks, and the companion paper extends it formally to world model planning traces [29].

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